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Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2018

Time: 03 Hours

Code: 011620 *Subject: Design of Steel Structure* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven): **[14]**

(a) In plastic analysis, the shape factor for circular sections is
(i) 1.5 (iii) 1.697
(ii) 1.6 (iv) None of the above

(b) A beam is defined as a structural member subjected to
(i) axial loading (iii) axial and transverse loading
(ii) transverse loading (iv) None of the above

(c) The failure of a web plate takes place by yielding if the ratio of the clear depth to thickness of the web is less than
(i) 45 (iii) 55
(ii) 60 (iv) 82

(d) The most economical section for a column is
(i) rectangular (iii) flat strip
(ii) solid round (iv) tubular section

(e) The distance between c.g. of compression and c.g. of tension flanges of a plate girder is known as
(i) overall depth (iii) effective depth
(ii) clear depth (iv) None of the above

(f) The allowable stress, in axial tension for rolled I-sections and channels, is taken as
(i) 1420 kg/cm^2 (iii) 2125 kg/cm^2
(ii) 1500 kg/cm^2 (iv) 1810 kg/cm^2

(g) If d is the distance between the flange angles, the vertical stiffeners in plate girders are spaced not greater than
(i) d (iii) 1.5d
(ii) 1.25d (iv) 1.75d

(h) The cross-section of a standard fillet is a triangle whose base angles are
(i) 45° and 45° (iii) 40° and 50°
(ii) 30° and 60° (iv) 20° and 70°

(i) A second horizontal stiffener is always placed at the neutral axis of the girder if the thickness of the web is less than
(i) *d*/250 for structural steel (iii) Both (i) and (ii)
(ii) *d*/225 for high tensile steel (iv) Neither (i) nor (ii)

(j) The thickness t of a single flat lacing should not be less than
(i) 1/30th length between inner end rivets (iii) 1/50th length between inner end rivets
(ii) 1/40th length between inner end rivets (iv) 1/60th length between inner end rivets

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Q.2 Explain the following: **[14]**

(a) Local and lateral buckling of beam

(b) Checks required for beam design

Q.3 Solve this question : **[14]**

Calculate the design compressive load for a column made up of ISHB 350 @ 710.2 N/m and 3.5 m high. The column is restrained in direction and position at both the ends. Use steel of grade Fe 410.

Q.4 Solve this question : **[14]**

Design a simply supported beam of span 4.2 m carrying reinforce concrete floor in which top compression flange is embedded. Beam is carrying 20 kN/m dead load and 20 kN/m imposed load, resume Fe 410 grade steel.

Q.5 Solve this question : **[14]**

Design a suitable angle section to carry tensile force of 250 kN. Use welded connection.

Q.6 Discuss the following: **[14]**

(a) Prying action

(b) Advantage of fillet weld over butt weld

(c) Comparison of welded joints with bolted joints

Q.7 Solve both questions : **[14]**

(a) Explain some of the common defects in the welds.

(b) Write the advantage of welded joints over bolted joints.

Q.8 Solve this question : **[14]**

Design a tension member to carry a pull of 830 kN. The member is 3.2 m between c/c of intersections. Design the member using channel section.

Q.9 Solve this question : **[14]**

A tie member of truss consists of double angle section each 80 mm x 80 mm x 8 mm welded on the opposite side of a 12 mm thick gusset plate. Design a fillet weld for making connection in the workshop. The factored tensile force in the member is 300 kN.

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